

12th March 2026 / CS6370

⇒ (contd.)

- SVD & rank reducⁿ

$$\|X\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n X_{ij}^2} \quad \left(\begin{array}{l} \text{goal is} \\ \text{to minimize} \\ \text{this.} \end{array} \right)$$

$$X = M - M_A$$

↳ discrepancy matrix

$\hat{M}$ is the best k -rank approximation to M in the least squares sense.

Quality of an approx. M is measured by Frobenius Norm.

- LSA & Orders of Co-occurrence
(First Order)

→ the words are related if they are present in the same doc.

(A, B) (A, C) (C, B)

(A, C), (C, D), (D, B)

second-order
co-occurrences
provides more
relatedness

- LSA can capture upto 5th order of co-occurrences.

- SVD for compression - as of original image, send a blurry image to reduce the dimensions

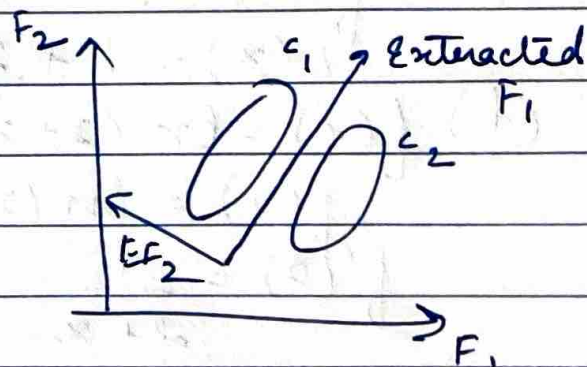
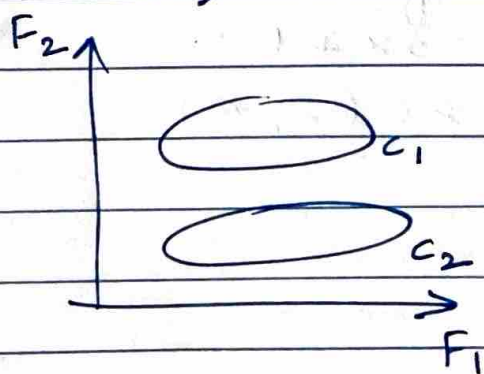
- mapping queries to new space

$$\hat{q} = q' U_K \Sigma_K^{-1}$$

- updated techniques $\rightarrow$ recomputaⁿ,
folding in, SVD update

- Is LSI good for classificaⁿ?

$\rightarrow$ some concepts might align with the clusters (agnostic to class labels $\rightarrow$ USL)



finds the closest approx. to the original dataset, to the minimal error.

- supervised SVD $\rightarrow$ classes act as additional features